Data Preparation

In addressing the task of preparing the provided data for analysis, several steps were undertaken to ensure the data was clean, consistent, and ready for further analysis. The data preparation process involved handling potential issues and errors in the dataset, including typos, extra whitespaces, sanity checks for impossible values, and missing values.

1. **Imported** necessary **libraries**.
2. **Loaded** the **CSV** data and conducted basic operations to assess the data's integrity.
3. **Identified missing values** using the isnull() function, calculating the sum of missing values per column.
   - Identified and located missing values using isnull() function by creating a boolean mask indicating whether each cell contains a missing value. We also checked the sum of missing values to count the number of missing values in each column.
   - To fix the missing values, we replaced:
   - Missing values in the rainfall column with its median to mitigate the impact of outliers and ensure the imputed values are representative of the data distribution.
   - Months and days column with its mode ensures that the most frequent values are used, maintaining the overall distribution and preserving the integrity of the data. This approach minimizes the distortion of the original data and helps maintain its statistical properties.
4. Utilized the describe() method to compute summary statistics and investigate outliers.
   - After checking the summary statistics, we found:
   - A minimum value of -10, which is physically impossible (negative).
   - A maximum value of 100000, which is an outlier with an extreme value.
5. **Addressed physically impossible values**, such as negative rainfall amounts and incorrect day or month entries.
   - Negative values were removed by filtering out rows where the value in the 'rainfall in mm' column is less than 0.
   - Outliers were removed from the 'Rainfall amount (millimeters)' column by filtering out values beyond 3 standard deviations from the mean.
6. **Checked for data inconsistency**, including textual representations of months and non-numeric day entries.
   - Invalid values were identified in the 'Month' and 'Day' columns by checking if they fell within expected ranges (1 to 12 for months and 1 to 31 for days). Rows with values outside the range were dropped.
   - To standardize the format of month names, we converted them to lowercase and created a dictionary to map month names to their corresponding numeric representations. After replacing month names with numeric values, we converted the 'Month' column to numeric datatype, filtering out rows with invalid month values falling outside the range of 1 to 12.
7. **Error against codebook**
   - Rows with years before 2013 or after 2023 were considered invalid. To correct this, we filtered the DataFrame to retain only the rows with years falling within the specified range of 2013 to 2023.

It's important to note that we chose to replace missing values with the median and mode instead of removing rows or imputing with 0 to preserve the integrity and statistical properties of the data. Additionally, we employed the calendar example to verify the correctness of dates and ensure consistency in the data.

From the output of print(df.info()), I found out that in our data

1. There are 3865 entries, ranging from index 0 to 3864.
2. Data type of each column
3. Missing values in each column

```
Data columns (total 4 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Year                            3865 non-null   int64
 1   Month                           3864 non-null   object
 2   Day                             3864 non-null   object
 3   Rainfall amount (millimetres)   3860 non-null   float64
dtypes: float64(1), int64(1), object(2)
memory usage: 120.9+ KB
```

Error 1 - Nan values - missing values

To identify and locate missing values, I used isnull() function and created a boolean mask.
To fix the missing values, I replaced missing values in

1. Rainfall column with its median - to mitigate the impact of outliers and ensure the imputed values are representative of the data distribution.
2. Months and days column with its mode - ensures that the most frequent values are used, maintaining the overall distribution and preserving the integrity of the data.

This approach minimizes the distortion of the original data and helps maintain its statistical properties.

Next, we checked the summary statistics using the describe() method.
Here we got to know that

1. Min value is -10, which is negative, which is not possible
2. Max value is 100000, which is an outlier with extreme value.

```
Summary Statistics:
               Year  Rainfall amount (millimetres)
count  3865.000000                    3860.000000
mean   2018.199224                      27.480829
std       3.064380                    1609.537526
min    2013.000000                     -10.000000
25%    2016.000000                       0.000000
50%    2018.000000                       0.000000
75%    2021.000000                       1.000000
max    2027.000000                  100000.000000
```

Error 2 - Negative value (-10) - Physically impossible values

To locate this error, we simply sorted rows where the value in column ‘rainfall in mm’ is less than 0.
Then, to mitigate this error, we simply removed all rows which have values less than 0, which was a data error.

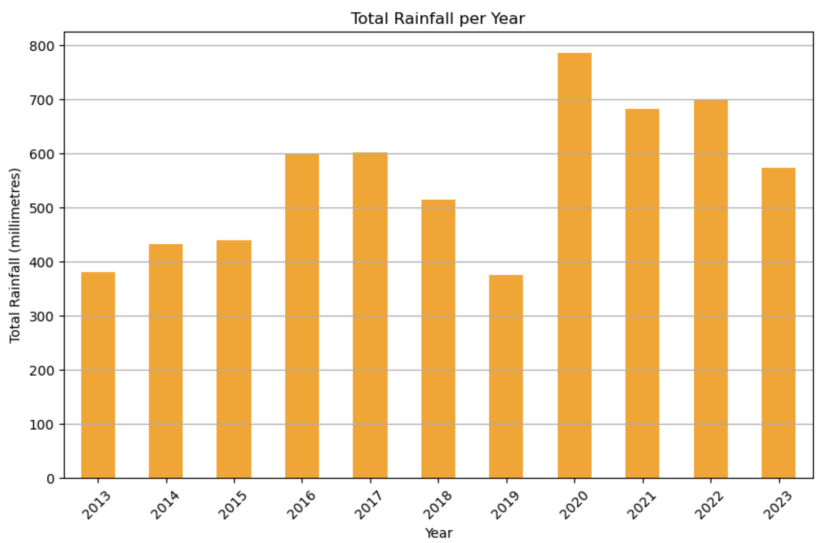
Error 3 - Outliers (extreme value 100000)

I removed outliers from the 'Rainfall amount (millimeters)' column by calculating the mean and standard deviation. Outliers, defined as values beyond 3 standard deviations from the mean, were filtered out. This

process helps ensure that extreme values do not disproportionately influence the analysis, maintaining the integrity of the dataset.

Error 4 - April Jan, 48, nine, 200 - Data inconsistency
I verified the validity of values in the 'Month' and 'Day' columns by checking if they fell within expected ranges. Any values outside the range of 1 to 12 for months and 1 to 31 for days were identified as invalid. This step ensures that only valid and meaningful data points are retained in the dataset for further analysis. To fix this, all the rows with values outside the range were dropped.
Instead of removing rows with months printed as text (April), instead of numbers (4), and to keep valid data, I standardized the format of month names by converting them to lowercase. Then, I created a dictionary to map month names to their corresponding numeric representations. After replacing month names with numeric values, I converted the 'Month' column to numeric datatype, filtering out rows with invalid month values falling outside the range of 1 to 12.

Error 5 - Error against codebook (year 2027)
I identified rows with years outside the specified range according to the codebook. Rows with years before 2013 or after 2023 were considered invalid. To correct this, I filtered the DataFrame to retain only the rows with years falling within the specified range of 2013 to 2023.



Data Exploration

Task 2.1: Highest Daily Rainfall in Each Month (2014) Analysis

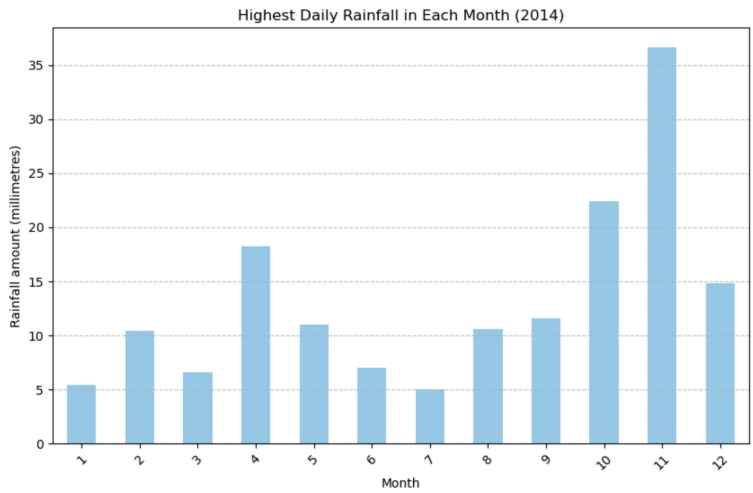
Introduction:
This analysis explores the highest daily rainfall recorded in each month of 2014 in ABC City. We prepared the data, identified the peak rainfall for each month, and visualized the results using a bar chart.

Data Preparation:
We loaded the dataset, filtered it for the year 2014, and pivoted the DataFrame to arrange data by days and months.

Exploration:
Grouping the data by month, we found the maximum daily rainfall for each month.

Visualization:
A bar chart was used to visualize the highest daily rainfall for each month, providing a clear representation of rainfall distribution throughout the year.

Conclusion:
The analysis offers insights into monthly rainfall patterns in 2014, aiding in understanding local precipitation trends.



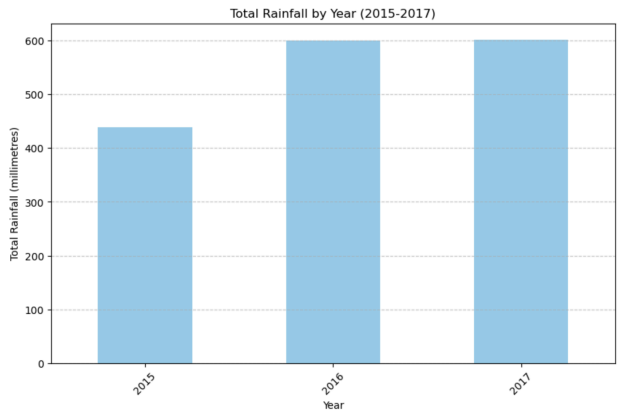
Task 2.2: Rainfall Data Analysis (2015-2017)

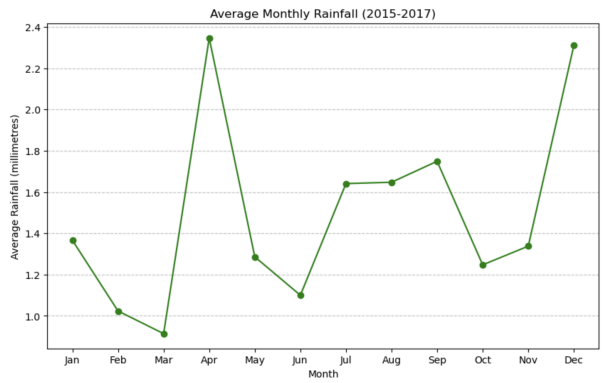
Introduction:
This report analyzes rainfall data from 2015 to 2017 in ABC City, examining yearly and monthly trends.

Yearly Analysis:
Total rainfall for each year was calculated and visualized using a bar chart, highlighting fluctuations in yearly rainfall.

Monthly Analysis:
Average monthly rainfall over the three-year period was computed and plotted, revealing variations across months.

Conclusion:
The analysis provides valuable insights into rainfall patterns, aiding in understanding seasonal trends.





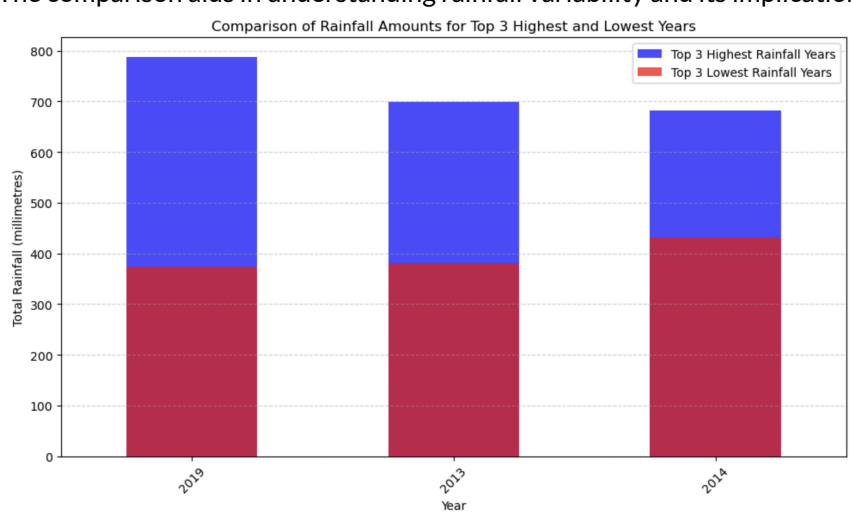
Task 2.3: Comparison of Rainfall Distribution for Top 3 Highest and Lowest Years

Introduction:
This analysis compares rainfall distribution for the top 3 highest and lowest years, offering insights into extreme weather events.

Analysis:
Top years with highest and lowest rainfall were identified, and a box plot was used to compare rainfall distribution.

Results:
The box plot illustrates differences in rainfall distribution between top years, informing about variability in rainfall patterns.

Conclusion:
The comparison aids in understanding rainfall variability and its implications for resource management.



Task 2.4: Monthly Rainfall Variation and Rolling Average Analysis

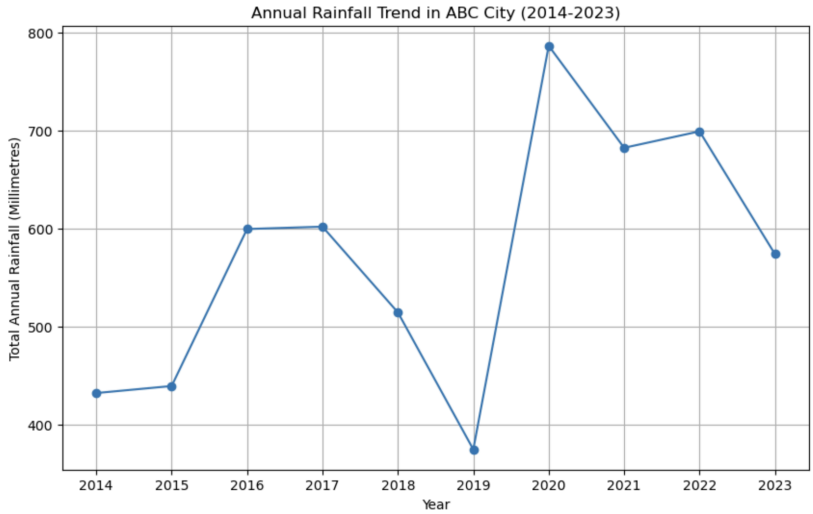
Introduction:
This analysis explores changes in rainfall patterns from 2014 to 2023 in ABC City, focusing on monthly variation and rolling averages.

Data and Methodology:
Filtered data for 2014-2023 was used to calculate total annual rainfall, visualize monthly variation, and compute rolling averages.

Findings:
Monthly variation plot reveals fluctuations in rainfall, while rolling average highlights long-term trends.

Conclusion:
The analysis provides insights into rainfall patterns over the decade, informing future planning and mitigation efforts.

Overall, the analyses offer comprehensive insights into rainfall patterns, supporting informed decision-making and resource management.



REFERENCES -

1. RMIT Canvas and course study material
2. https://docs.python.org/3/library/index.html